

AE 2610, Lab #2

Aerodynamic Loads

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Abstract

This experiment investigated the aerodynamic forces on a symmetric rectangular wing in a subsonic and low turbulence wind tunnel. By combining a six-component load cell and a Baratron pressure transducer, it was possible to measure the x-, y-, and z- components of both force and torque on the wing as well as the Baratron voltage data. The angle of attack of the wing, both with and without an end plate, varied from 0 to 22 degrees in high (~16.5 m/s) and low (~12.5 m/s) speed environments. The objective of this lab was to reduce the data and solve for non-dimensional coefficients (Lift, Drag, Moment) before evaluating the impact of varying wind speed and end plates on wing performance. The results show that both lift and drag increase with the angle of attack until stall. With stall, every configuration experienced a significant drop in lift, spike in drag, and reversal in moment coefficient. Additionally, it was found that a faster air flow notably delayed the critical stall angle from roughly 12° to 19°. End plates effectively produced a higher maximum lift coefficient before stalling. Overall, the error of this experiment was minimized through repetition, and the results support established aerodynamic findings.

Introduction

The goal of this lab was to reduce the raw data gathered and calculate the non-dimensional lift, drag, and pitching moment coefficients across different angles of attack in the four main configurations (High vs. Low Speed, With vs. Without End Plate). Additionally, this lab aimed to assess the impact of end plates by comparing the results from the different configurations to better understand how a physical barrier at the tip of a wing can influence important aerodynamic characteristic like lift coefficient and induced drag as well as the overall efficiency of a wing in a wind tunnel.

Wind tunnels and experimental testing are a great way to test and understand the ability of a wing to generate lift and drag, especially at higher angles of attack where the air flow separates. Measurements were taken in the test section of a subsonic, low-turbulence wind tunnel, which was able to draw air from the room and convert it into a steady, uniform flow over the stationary wing model. The symmetric wing was mounted vertically on a strain-gauge load cell at its quarter-chord line, which featured six components to accurately convert aerodynamic loads into voltage outputs for each of the force and torque components. Additionally, a stepper motor was responsible for moving the wing through a range of angles of attack, pausing at each increment so the data acquisition system can log simultaneous load cell and Baratron readings.

Experimental Setup

This experiment was conducted in an open-return Low Turbulence Wind Tunnel (LTWT) in Montgomery Knight 104. This lab revolved around a symmetric and rectangular model wing mounted vertically in the wind tunnel test section. Additionally, it is important to note that this model wing was evaluated both with and without a removable end plate. Ultimately, the following laboratory setup enabled accurate and repeatable measurements of aerodynamic forces and moments on the model wing.

Lab Instruments & Connections

FlowKinetics 3DP1A Manometer/Flow Speed Meter

- Responsible for measuring the local ambient pressure and temperature. Using information from the FlowKinetics manometer, it was possible to calculate both air density and flow speed. The FlowKinetics manometer served as an independent reference in this lab, meaning that it did not produce any new signals throughout the course of the experiment.

Load Cell (ATI Gamma SI-130-10)

- The ATI Gamma SI-130-10 is a six-component load cell that can record data on all three force components and all three moment components at the same time. It is able to work by combining an array of strain gauges bonded to a steel block and designed in a way to produce both distinct and measurable strain outputs. The ATI Gamma SI-130-10 was positioned just below the wind tunnel floor and mounted directly on the wing's quarter-chord line.

Load Cell (Data Acquisition System)

- PC responsible for running LabVIEW VI. This computer is interfaced with the Baratron, load cell, Baratron, and stepper motor in order to increase automation and log the resulting voltage outputs.

Load Cell (Stepper Motor & ESP32 Microcontroller)

- The stepper motor was responsible for turning the wing precisely in order to make measurements at different angles of attack. This stepper motor was controlled by an ESP32 microcontroller, which was connected to a PC via USB. The ESP32 microcontroller was ultimately responsible for precisely varying the wing's angle of attack.

Low Turbulence Wind Tunnel

- Responsible for directing air through a controlled duct in order to simulate the aerodynamic conditions experienced by a moving object. Using similarity parameters like Mach Number and Reynold's Number, wind tunnels enable users to interpret data and apply their findings to a full-sized vehicle. Air is drawn from the room and first passes through a settling chamber that contains both a honeycomb to remove swirl and screens to break down any eddies in the flow. Then, the air flow increases as the wind tunnel area decreases and enters a constant-area test section maintaining constant velocity and pressure. Finally, air flows through the exhaust of the wind tunnel and recirculates back into the room.

Pitot-Static Probe & Baratron

- A pitot-static probe was mounted in the freestream of the test section in order to properly measure dynamic pressure. The pitot-static probe worked by utilizing two main holes with one facing directly into the wind tunnel flow and the other oriented in a way that enables air to flow along its surface. The first hole was responsible for recording the stagnation pressure while the second hole was responsible for recording static pressure. The two holes were connected to the two sides of a Baratron via long tubing. A Baratron is a capacitance-type transducer that could interpret difference in pressure measured in the pitot-static probe as a voltage change. This voltage change was then recorded by the data acquisition system in order to determine the dynamic pressure of the freestream wind flow as well as the wind tunnel velocity.

Measurement Procedure

The experiment began with powering on and initializing the Low Turbulence Wind Tunnel and all necessary instrumentation. The symmetric wing model was already mounted vertically on the six-component strain-gauge load cell and positioned above the stepper motor. After completing the safety checklist, the LabVIEW VI was launched and communication with the stepper controller was confirmed. Before any airflow was introduced, the ambient room pressure and temperature were recorded using a FlowKinetics manometer. This information will be important for later experimental calculations. The instrumentation was then properly calibrated. The Baratron pressure transducer and the load cell were zeroed. Additionally, the model wing was aligned until the measured lift read zero.

With the model properly aligned, main data could begin. The data collection portion of this lab consisted of changing the angle-of-attack at two different freestream velocities. To begin, the fan speed was set to produce a high-speed flow of about 16.5 m/s, and the lab group manually rotated the wing with the end plate from 0° to 22° in 1° increments. At each angle, the lab group paused the system for around three seconds while the load cell and Baratron data were logged simultaneously at 1000 samples per second. After completing the high-speed sweep, the wing was returned to its original position, and the tunnel speed was reduced to roughly 12.5 m/s, and a second sweep from 0° to 20° was performed using the same procedures.

Following the first two experiments, the configuration of the wing was then modified to remove the end plate. The lab group powered down the wind tunnel, and the test section was opened to remove the aluminum end plate, once the fan had stopped rotating. After closing the test section and getting back to safety, the tunnel was restarted and the lab group created a new Excel data file and zeroed the sensors again. The full high- and low-speed procedures were used once again for this new configuration. Finally, the raw .csv files were exported and checked before the wind tunnel was shut down.

Results and Discussion

Initial Conditions & Wing Characteristics

To start this lab, simply make sure to identify the initial conditions of the environment surrounding the wind tunnel. The air density and temperature were found to be the constant in the room for all four configurations, being identified as 1.15 kg/m^3 and 21.1°C or 294.25 K

respectfully. Using the FlowKinetics Manometer, the fast air speed and the slow air speed were also recorded to 16.0 m/s and 12.0 m/s respectively, but this information was only meant to be used as a logic check. The actual air speeds of the four configurations were calculated later in the experiment using dynamic pressure and ambient air density. Finally, the ambient pressure, recorded using the FlowKinetics Manometer, came out to be 98.08 kPa for the experiments with the end plate and 98.02 kPa for the experiments without the end plate.

The characteristic geometry of the wing and the end plate used in the test section of the wind tunnel during this experiment can be found in Table I. This symmetric wing has a span of 0.686 m and a chord of 0.310 m, giving a total effective planform area of 0.213 m². This relationship can be seen in the expression below:

$$S = b \times c$$

The span and chord of the wing are extremely important later on because they are used for all non-dimensional coefficients, like the coefficient of lift, drag, and moment. It is also important to know that the airfoil was considerably thick with a maximum thickness of 0.076 m and a wing thickness ratio of 0.246. This relationship can be seen in the expression below where TR represents the thickness ratio, t_{max} represents the maximum physical thickness of the wing, and c represents the chord length.

$$TR = \frac{t_{max}}{c}$$

Additionally, Table I also lists many of the dimensions of the removable aluminum end plate. It is important to note that the end plate used in this experiment had a chord of 0.559 m and a height of 0.229 m, making its area substantially larger than that of the wing's planform area.

This size enables the end plate to efficiently act as physical barrier and suppress the formation of tip vortexes during testing.

Table I. The characteristic geometry of the wing and the plate used in the test section.

Wing Characteristic	Imperial Units	Metric Units
Chord of Wing	12.2 in	0.310 m
Chord of End Plate	22 in	0.559 m
Span (Wing)	27.0 in	0.686 m
Span (End Plate)	0.08 in	0.002 m
Max Thickness (Wing)	3.0 in	0.076 m
Max Thickness (End Plate)	9.0 in	0.229 m
Planform	329.4 in ²	0.213 m ²
Thickness Ratio	0.246 (Unitless)	0.246 (Unitless)

Raw Data, Averages, and Variances

With the wing characteristics and the initial conditions recorded, the experiment was ready to commence. Following the procedures in the previous section, the lab group carefully went step-by-step and recorded all the force, moment, and Baratron voltage data using the combination of the load cell and the data acquisition system. Afterwards, the group uploaded thousands of data points collected to Excel for further reduction. More specifically, the data points were broken down into two sections, one including data collected from the wing with the end plate and the other including the data collected from the wing without the end plate.

As for the additional reduction of this data, these measurements could be further broken down and separated into two groups of two, based on end plate inclusion and wind speed. Once the data was properly organized, the first task was to find the average/mean of all the data points. This task is best completed through a simple MATLAB code that implements the following equation for each component of force and torque as well as the Baratron voltage measurements.

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

Utilizing the same MATLAB code as before, the next task was to calculate the variance for each component of force and torque as well as the Baratron voltage measurements. The variance of a data point simply refers to the average spread of the thousands of data points. This characteristic must be recorded in the MATLAB script and can be determined by the following equation:

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$

The following tables were derived from the logic above and utilized a MATLAB script in order to effectively read the thousands of data points and compute the reduced values.

Table II. Averages and variances of wing at high speed with plate.

Alpha_deg	Mean_Fx	Var_Fx	Mean_Fy	Var_Fy	Mean_Fz	Var_Fz	Mean_Tx	Var_Tx	Mean_Ty	Var_Ty	Mean_Tz	Var_Tz	Mean_Bar	Var_Bar	atr
0	-0.34499	0.066209	1.124414	0.021905	0.495235	0.017037	-0.51052	0.000221	-0.10724	0.003067	0.040657	0.000133	1.131276	0.000103	
1.008	-2.25099	0.059191	1.157527	0.011543	0.223232	0.015289	-0.51442	0.000331	-0.84403	0.002072	0.004336	9.70E-05	1.129186	3.99E-05	
2.016	-4.62696	0.055254	1.151476	0.009759	-0.11901	0.014607	-0.49993	0.000583	-1.75381	0.004202	-0.03594	0.000157	1.134745	1.85E-05	
3.024	-6.68871	0.048067	1.069307	0.008422	-0.4141	0.015447	-0.46127	0.000214	-2.5408	0.003308	-0.06779	0.00018	1.13247	8.95E-05	
4.032	-8.92653	0.056859	0.943154	0.011026	-0.71352	0.018288	-0.40902	0.000303	-3.39362	0.00359	-0.10012	0.00015	1.137182	3.33E-05	
5.04	-11.0996	0.032217	0.757119	0.007279	-1.02431	0.017174	-0.33648	0.000354	-4.25413	0.002471	-0.13759	0.000156	1.124019	2.28E-05	
6.048	-13.141	0.073044	0.512939	0.01023	-1.27272	0.013664	-0.23648	0.000822	-5.04603	0.008167	-0.16517	0.000209	1.123495	5.28E-05	
7.056	-15.3665	0.039166	0.226429	0.013881	-1.54181	0.017223	-0.11926	0.001541	-5.9155	0.003329	-0.19765	0.000244	1.128204	2.88E-05	
8.064	-17.3778	0.043472	-0.11718	0.007406	-1.71585	0.013943	0.017308	0.000462	-6.71035	0.003464	-0.22767	0.000226	1.126	0.000198	
9.072	-19.6237	0.093168	-0.53014	0.010714	-1.89004	0.019811	0.185786	0.001357	-7.61031	0.015531	-0.25896	0.000404	1.125312	8.32E-05	
10.08	-21.1674	0.092348	-0.91314	0.011057	-1.97819	0.014852	0.343694	0.001006	-8.2438	0.014703	-0.28861	0.000389	1.130036	2.30E-05	
11.088	-23.1638	0.088124	-1.39773	0.010307	-2.09385	0.018085	0.538918	0.001125	-9.03816	0.012694	-0.31048	0.000469	1.133164	0.000131	
12.096	-25.1967	0.053536	-1.93879	0.019315	-2.16423	0.018926	0.756251	0.002344	-9.84119	0.006386	-0.32726	0.000222	1.123123	2.59E-05	
13.104	-27.0302	0.246418	-2.52479	0.022843	-2.18336	0.015562	0.984504	0.003955	-10.564	0.046604	-0.3441	0.001058	1.130752	5.06E-05	
14.112	-28.8643	0.141669	-3.1921	0.044761	-2.19759	0.017242	1.251751	0.008728	-11.2912	0.025324	-0.36771	0.000684	1.139125	3.05E-05	
15.12	-30.4383	0.189621	-3.82693	0.097155	-2.20096	0.023232	1.498516	0.018746	-11.894	0.035621	-0.38348	0.000824	1.11826	2.81E-05	
16.128	-32.0055	0.065331	-4.53398	0.046565	-2.15411	0.029873	1.768251	0.008363	-12.4798	0.009305	-0.40297	0.000449	1.133533	2.14E-05	
17.136	-33.384	0.356462	-5.10182	0.040451	0.683449	0.103401	2.00085	0.010813	-12.8838	0.027996	-0.41235	0.001486	1.146739	2.70E-05	
18.144	-34.2293	0.215782	-5.54793	0.028312	3.865952	14.59173	2.169636	0.003926	-13.0208	0.004552	-0.39603	0.000847	1.145099	3.79E-05	
19.152	-15.024	0.885171	4.837797	0.040854	0.377064	0.065199	-2.15314	0.008672	-4.90104	0.168539	0.43295	0.00518	1.171582	1.97E-05	
20.16	-15.8365	1.022273	4.86467	0.039544	0.481291	0.077078	-2.14308	0.007409	-5.26578	0.212162	0.451621	0.00483	1.184155	6.22E-05	
21.168	-16.7024	0.356289	4.823674	0.163487	0.582921	0.049556	-2.10542	0.033601	-5.58568	0.066715	0.475184	0.002324	1.189834	2.75E-05	
22.176	-17.1079	1.239875	4.824542	0.062948	0.820933	0.061357	-2.08103	0.010932	-5.79685	0.250015	0.490321	0.006343	1.191688	7.25E-05	

Table III. Average and variance of wing at low speed with plate.

Alpha_deg	Mean_Fx	Var_Fx	Mean_Fy	Var_Fy	Mean_Fz	Var_Fz	Mean_Tx	Var_Tx	Mean_Ty	Var_Ty	Mean_Tz	Var_Tz	Mean_Bar	Var_Bar	atr
0	-0.09355	0.032558	0.580502	0.004557	0.385747	0.007606	-0.30645	0.000113	-0.0274	0.003399	0.026053	0.000123	0.640587	3.29E-05	
1.008	-1.18395	0.016812	0.599677	0.004791	0.235689	0.006831	-0.307	0.000183	-0.44661	0.000756	0.004811	5.77E-05	0.633555	1.99E-05	
2.016	-2.49519	0.017267	0.606127	0.004123	0.044565	0.007133	-0.30175	0.000165	-0.94599	0.001113	-0.01717	7.43E-05	0.633364	4.22E-05	
3.024	-3.70255	0.018287	0.566165	0.004395	-0.13076	0.006869	-0.28219	0.000159	-1.40488	0.001453	-0.03654	0.000117	0.629384	1.81E-05	
4.032	-4.972	0.026054	0.505384	0.004089	-0.31752	0.00703	-0.25481	0.000129	-1.88504	0.002571	-0.05441	0.000111	0.632967	1.59E-05	
5.04	-6.10813	0.02209	0.405504	0.003858	-0.50692	0.006811	-0.21602	0.00035	-2.33528	0.002465	-0.0761	9.26E-05	0.631893	1.90E-05	
6.048	-7.29213	0.017743	0.290363	0.003232	-0.66718	0.007056	-0.16612	0.000125	-2.79127	0.00171	-0.0927	0.000125	0.632284	1.90E-05	
7.056	-8.45804	0.031741	0.15182	0.003418	-0.8197	0.008014	-0.10645	8.81E-05	-3.24857	0.004206	-0.1124	0.000162	0.633163	2.79E-05	
8.064	-9.42449	0.021764	-0.02089	0.004837	-0.95006	0.007792	-0.03486	0.000516	-3.63154	0.002091	-0.13141	0.000106	0.630208	1.99E-05	
9.072	-10.718	0.027564	-0.2238	0.008888	-1.09376	0.007461	0.050924	0.001358	-4.14295	0.003846	-0.15251	0.000147	0.635725	2.89E-05	
10.08	-11.6791	0.020719	-0.42649	0.004211	-1.2069	0.007208	0.133954	0.000364	-4.52134	0.001967	-0.16631	0.000179	0.640326	2.36E-05	
11.088	-12.5818	0.029938	-0.63434	0.003688	-1.28777	0.008126	0.222872	0.000172	-4.87652	0.003083	-0.17558	0.000167	0.64205	2.66E-05	
12.096	-5.26348	2.018109	2.572729	0.074194	-0.08096	0.2147	-1.23387	0.012978	-1.62448	0.40925	0.240089	0.010458	0.651257	1.49E-05	
13.104	-5.49799	0.534282	2.636284	0.060801	-0.04167	0.186816	-1.25707	0.008139	-1.69627	0.109797	0.228036	0.002739	0.656103	2.02E-05	
14.112	-6.35198	0.439304	2.664086	0.055421	-0.08066	0.126509	-1.27084	0.008393	-1.99388	0.086047	0.240792	0.002066	0.659149	1.43E-05	
15.12	-6.80956	0.777187	2.647717	0.023492	-0.03055	0.067509	-1.26164	0.002999	-2.14552	0.147821	0.231999	0.003929	0.660181	3.65E-05	
16.128	-7.15647	0.458018	2.637465	0.062519	0.065285	0.067042	-1.25035	0.011429	-2.2431	0.087973	0.22293	0.002629	0.663172	1.42E-05	
17.136	-7.5348	0.523459	2.696051	0.030742	0.072431	0.059194	-1.26085	0.005509	-2.42883	0.104966	0.232265	0.002687	0.668099	2.52E-05	
18.144	-7.73433	0.891972	2.686269	0.056125	0.145904	0.05794	-1.24489	0.012078	-2.52587	0.180545	0.225113	0.003722	0.661398	1.71E-05	
19.152	-8.04717	0.219468	2.73357	0.028884	0.25312	0.032451	-1.24994	0.00457	-2.65044	0.036908	0.227935	0.001026	0.668239	1.48E-05	
20.16	-8.70561	0.804109	2.711882	0.029402	0.30548	0.028544	-1.23355	0.005681	-2.8907	0.16387	0.24646	0.004256	0.678812	1.85E-05	

Table IV. Average and variance of wing at high speed without plate.

Alpha_deg	Mean_Fx	Var_Fx	Mean_Fy	Var_Fy	Mean_Fz	Var_Fz	Mean_Tx	Var_Tx	Mean_Ty	Var_Ty	Mean_Tz	Var_Tz	Mean_Bar	Var_Bar	Baratr
0	-0.04285	0.047606	1.098473	0.012922	0.864232	0.01858	-0.34985	0.000817	-0.01625	0.001073	0.044828	0.000164	1.133057	3.67E-05	
1.008	-1.77668	0.050166	1.126652	0.015979	0.635899	0.02176	-0.35206	0.000609	-0.63965	0.001591	0.009312	0.000183	1.119649	5.81E-05	
2.016	-3.81855	0.049583	1.121175	0.010765	0.378547	0.015527	-0.3417	0.00044	-1.37291	0.002448	-0.02972	0.000125	1.127494	6.56E-05	
3.024	-5.68613	0.044118	1.074605	0.013288	0.158336	0.019592	-0.31961	0.00046	-2.0401	0.001728	-0.06158	0.00014	1.137886	2.95E-05	
4.032	-7.59606	0.043986	0.975837	0.012775	-0.07621	0.03158	-0.28254	0.00117	-2.72742	0.003013	-0.09342	0.000188	1.120756	2.20E-05	
5.04	-9.53619	0.033213	0.824613	0.007631	-0.31549	0.020824	-0.23327	0.000525	-3.44518	0.00148	-0.12681	0.000116	1.130064	4.01E-05	
6.048	-11.4704	0.083677	0.636621	0.010956	-0.50983	0.025767	-0.16333	0.000869	-4.15381	0.010513	-0.15346	0.000317	1.13137	4.37E-05	
7.056	-13.4302	0.037054	0.409989	0.005197	-0.69465	0.018396	-0.08007	0.000221	-4.87193	0.002081	-0.18021	0.000207	1.124238	3.65E-05	
8.064	-15.1505	0.072845	0.139115	0.021785	-0.85911	0.015161	0.018764	0.003372	-5.52069	0.008839	-0.20379	0.000311	1.130723	5.87E-05	
9.072	-17.0232	0.071616	-0.18857	0.007343	-1.00052	0.017075	0.139589	0.000438	-6.23169	0.008024	-0.2277	0.00031	1.125574	3.18E-05	
10.08	-18.9629	0.093205	-0.53714	0.007303	-1.11343	0.018044	0.267819	0.000255	-6.96199	0.00974	-0.24971	0.000438	1.132231	5.48E-05	
11.088	-20.6314	0.10536	-0.91407	0.007148	-1.19651	0.017572	0.410083	0.000458	-7.61731	0.012788	-0.26972	0.000479	1.132673	2.89E-05	
12.096	-22.2355	0.100606	-1.34553	0.009646	-1.28015	0.022724	0.565789	0.001111	-8.22739	0.013028	-0.2766	0.000407	1.126489	0.000117	
13.104	-24.2665	0.100256	-1.83959	0.009398	-1.32982	0.025478	0.744898	0.000594	-9.00117	0.01269	-0.28837	0.000472	1.132189	8.04E-05	
14.112	-25.9741	0.066572	-2.37465	0.008898	-1.38309	0.023705	0.934211	0.000557	-9.65023	0.006864	-0.29528	0.000308	1.128692	4.67E-05	
15.12	-27.5949	0.15797	-2.91822	0.011181	-1.39588	0.029658	1.127533	0.000957	-10.27	0.021514	-0.3057	0.000639	1.141016	0.000178	
16.128	-29.4294	0.123479	-3.56939	0.011294	-1.39982	0.01789	1.356645	0.00106	-10.9648	0.018149	-0.3183	0.000558	1.138762	7.97E-05	
17.136	-30.7373	0.10817	-4.11556	0.009801	-1.35738	0.023598	1.549523	0.001182	-11.4544	0.015766	-0.31879	0.00056	1.145797	3.46E-05	
18.144	-31.5904	0.275738	-4.54552	0.029867	-1.31364	0.024242	1.695779	0.004369	-11.7528	0.04328	-0.29257	0.001369	1.131387	3.98E-05	
19.152	-15.2305	1.770086	4.709629	0.331616	-0.15123	0.053781	-1.65805	0.057739	-5.0855	0.289569	0.497312	0.010267	1.174906	7.22E-05	
20.16	-16.7854	1.521379	4.675206	0.055677	-0.27675	0.039275	-1.65033	0.008123	-5.59839	0.288966	0.541763	0.009491	1.175884	7.27E-05	
21.168	-17.1381	1.092876	4.678017	0.21517	-0.29596	0.031263	-1.64273	0.038557	-5.68549	0.184925	0.532704	0.00472	1.184451	4.14E-05	
22.176	-18.1168	1.071623	4.636076	0.206907	-0.38515	0.02832	-1.62165	0.035785	-6.04211	0.16181	0.564795	0.003926	1.180622	8.42E-05	

Table V. Average and variance of wing at low speed without plate.

Alpha_deg	Mean_Fx	Var_Fx	Mean_Fy	Var_Fy	Mean_Fz	Var_Fz	Mean_Tx	Var_Tx	Mean_Ty	Var_Ty	Mean_Tz	Var_Tz	Mean_Bar	Var_Bar	Baratr
0	-0.04727	0.018635	0.60471	0.006119	0.664526	0.007426	-0.20884	0.000198	-0.01224	0.000808	0.024155	0.000141	0.631717	2.21E-05	
1.008	-0.96387	0.018178	0.615886	0.004404	0.543181	0.008472	-0.20916	0.000221	-0.34354	0.000875	0.004154	6.95E-05	0.627893	2.70E-05	
2.016	-2.10142	0.015058	0.625245	0.004561	0.397435	0.007226	-0.20634	0.000322	-0.74982	0.000747	-0.01743	8.61E-05	0.633571	1.75E-05	
3.024	-3.16681	0.015823	0.610326	0.007703	0.260084	0.007237	-0.19747	0.000658	-1.12968	0.000833	-0.03457	8.06E-05	0.635634	2.74E-05	
4.032	-4.25761	0.01671	0.555557	0.004032	0.116471	0.006629	-0.17588	0.000247	-1.51772	0.00161	-0.05096	8.97E-05	0.630451	1.19E-05	
5.04	-5.31139	0.01355	0.485987	0.003807	-0.03481	0.007255	-0.15282	0.000324	-1.90892	0.000905	-0.07113	6.73E-05	0.640445	2.55E-05	
6.048	-6.36665	0.017886	0.39306	0.003885	-0.17479	0.007084	-0.11472	0.000233	-2.29244	0.002027	-0.08569	0.000164	0.633084	1.66E-05	
7.056	-7.38632	0.021423	0.274415	0.003871	-0.30288	0.00777	-0.06906	0.000378	-2.67117	0.002174	-0.10348	0.000137	0.628936	2.55E-05	
8.064	-8.37921	0.021928	0.140332	0.004579	-0.41306	0.0089	-0.01698	0.000424	-3.04166	0.001692	-0.11955	0.000179	0.633431	3.62E-05	
9.072	-9.36432	0.021731	-0.01396	0.007315	-0.49531	0.007843	0.041048	0.000949	-3.4201	0.001746	-0.13714	0.000142	0.635086	2.62E-05	
10.08	-10.3343	0.027538	-0.17786	0.004908	-0.58477	0.00814	0.104259	0.00027	-3.7824	0.003053	-0.14919	0.000201	0.634603	1.45E-05	
11.088	-11.1398	0.079732	-0.33892	0.00672	-0.65369	0.00864	0.166761	0.000775	-4.08911	0.011596	-0.15622	0.000378	0.630699	1.64E-05	
12.096	-5.78042	0.53254	2.518435	0.047715	0.195678	0.017817	-0.90345	0.008669	-2.0369	0.091816	0.283189	0.002373	0.65305	2.22E-05	
13.104	-6.5516	0.373493	2.584431	0.014673	0.116175	0.015682	-0.92424	0.002312	-2.3261	0.063018	0.307461	0.002938	0.655866	2.05E-05	
14.112	-6.61073	0.449905	2.632809	0.012268	0.121407	0.016092	-0.94959	0.002033	-2.23617	0.080207	0.282006	0.00332	0.659381	1.38E-05	
15.12	-7.04136	0.296525	2.63062	0.016419	0.08331	0.014299	-0.94764	0.002699	-2.36705	0.055317	0.278535	0.002108	0.659779	2.49E-05	
16.128	-7.74284	0.45644	2.650615	0.061726	0.018257	0.017647	-0.95035	0.010714	-2.61475	0.078295	0.289794	0.002807	0.661917	3.18E-05	
17.136	-7.96239	0.795026	2.679324	0.03627	-0.00236	0.023346	-0.95883	0.006418	-2.67966	0.139536	0.283084	0.003494	0.662672	2.14E-05	
18.144	-8.40994	0.674157	2.661114	0.067304	-0.04379	0.020819	-0.94888	0.012021	-2.85827	0.119273	0.291749	0.003535	0.663408	1.96E-05	
19.152	-8.78863	0.396705	2.713544	0.009825	-0.07078	0.016471	-0.96639	0.00174	-2.9515	0.064816	0.288051	0.002105	0.67432	1.55E-05	
20.16	-9.13826	0.334806	2.70767	0.041321	-0.09439	0.013716	-0.95983	0.006985	-3.05656	0.054541	0.287427	0.00162	0.671821	2.05E-05	

Derived Data

First, this experiment required the derived data for dynamic pressure. The derivation for dynamic pressure data was quite simple and revolved around the average Baratron voltage reading and the

Baratron sensitivity. It is important to note that in order for this derivation to be successful, the Baratron sensitivity must be converted from mmHg/Volt to Pa/Volt because the ambient pressure reading was given in kPa. The following equations describing this relationship must be utilized in MATLAB and can be found below:

$$1.016 \text{ mmHg/Volt} \times 133.322 \text{ Pa/mmHg} = 135.455 \text{ Pa/Volt}$$

$$q_{\infty} = \text{Baratron Voltage} \times 135.46 \text{ Pa/Volt}$$

Second, this experiment required the derived data for real air flow velocity. Once all the data values for the reduced dynamic pressure were collected, it became possible to solve for the real air speed in the wind tunnel. By combining the reduced dynamic pressure q and the air density ρ , simply implement the following equation in MATLAB and record the new data points in a new column:

$$v_{\infty} = \sqrt{\frac{2q_{\infty}}{\rho}}$$

In this experiment, the real air flow velocity slightly varied depending across the different configurations, but the high-speed average turned out to be 16.5 m/s whereas the low-speed average turned out to be 12.5 m/s.

Third, this experiment required the derived data for drag and lift. This derivation is extremely important because the sensors on the load cell change according to the direction of the wing and thus the angle of attack. The only two components that contribute to lift and drag are the x- and y- force components acting on the wing. This relationship between the two force components as well as the angle of attack must be included in the derivation of lift and drag and can be implemented in either MATLAB or Excel using the following two equations:

$$L = F_x \cos(\alpha) - F_y \sin(\alpha)$$

$$D = F_y \cos(\alpha) + F_x \sin(\alpha)$$

Finally, this experiment required the derived data for the coefficients of lift, drag, and moment. This is the data that the experiment is centered around in order to see how the four different configurations impact the aerodynamic abilities and efficiency of the wing. To derive the coefficient of lift and the coefficient of drag simply take the lift/drag at an angle and divide it by the product of dynamic pressure and planform area. To derive the moment coefficient simply take the moment, which can be represented by the z-component of the torque, and divide it by the product of the dynamic pressure, planform area, and chord length. All these equations can easily be implemented in either MATLAB or Excel and can be found below:

$$C_L = \frac{L}{q_\infty S}$$

$$C_D = \frac{D}{q_\infty S}$$

$$C_M = \frac{M}{q_\infty S c}$$

The accumulation of these four steps represents the entirety of what is asked in the derived data section. In order to better summarize and portray this data, the following tables are able to provide all the specific calculations made using the logic above. These tables are also divided up based on wind speed and the inclusion of the end plate or not.

Table VI. Derived data of wing at high speed with plate.

Alpha deg	DynPressure Pa	TrueAirspeed m/s	Lift N	Drag N	CL	CD	CM
0	153.2376	16.32483	0.344993	1.124414	0.010594	0.034528	0.004029
1.008	152.9545	16.30975	2.230278	1.196947	0.068613	0.036823	0.00043
2.016	153.7075	16.34984	4.583589	1.313533	0.14032	0.040212	-0.00355
3.024	153.3993	16.33344	6.622981	1.420676	0.20316	0.043579	-0.00671
4.032	154.0376	16.36739	8.838122	1.568476	0.269987	0.047914	-0.00987
5.04	152.2546	16.27239	10.99021	1.729308	0.33966	0.053445	-0.01372
6.048	152.1837	16.2686	13.0138	1.894639	0.402388	0.058582	-0.01648
7.056	152.8214	16.30265	15.22232	2.112328	0.468712	0.065041	-0.01964
8.064	152.5229	16.28672	17.22245	2.321731	0.531336	0.071629	-0.02267
9.072	152.4298	16.28175	19.46184	2.570675	0.60079	0.079357	-0.0258
10.08	153.0697	16.31589	21.00049	2.805737	0.645579	0.086252	-0.02863
11.088	153.4933	16.33845	23.00025	3.083154	0.705102	0.094518	-0.03072
12.096	152.1333	16.2659	25.04352	3.384221	0.774605	0.104675	-0.03267
13.104	153.1666	16.32105	26.89871	3.669212	0.826374	0.112724	-0.03411
14.112	154.3008	16.38137	28.77151	3.941876	0.877412	0.120211	-0.03619
15.12	151.4745	16.23065	30.38279	4.245123	0.943837	0.131874	-0.03844
16.128	153.5433	16.34111	32.00538	4.535091	0.980846	0.138984	-0.03985
17.136	155.3322	16.43602	33.40521	4.960945	1.011956	0.150284	-0.04031
18.144	155.11	16.42427	34.25496	5.38715	1.039184	0.163429	-0.03877
19.152	158.6972	16.6131	12.6053	9.499044	0.37376	0.281656	0.041427
20.16	160.4003	16.70201	13.18969	10.02457	0.386935	0.294083	0.042755
21.168	161.1696	16.74201	13.83361	10.52952	0.403888	0.307422	0.044771
22.176	161.4207	16.75505	14.02139	10.9251	0.408734	0.318475	0.046125

Table VII. Derived data of wing at low speed with plate.

Alpha deg	DynPressure Pa	TrueAirspeed m/s	Lift N	Drag N	CL	CD	CM
0	86.77111	12.28439	0.093549	0.580502	0.005073	0.03148	0.004559
1.008	85.81852	12.21678	1.173217	0.620413	0.064329	0.034018	0.000851
2.016	85.79262	12.21493	2.472325	0.693529	0.135602	0.038039	-0.00304
3.024	85.25355	12.1765	3.66753	0.760702	0.202428	0.041987	-0.00651
4.032	85.73883	12.2111	4.924156	0.853732	0.270248	0.046855	-0.00964
5.04	85.59341	12.20074	6.04889	0.940542	0.33254	0.051707	-0.0135
6.048	85.64635	12.20451	7.220947	1.057057	0.396729	0.058076	-0.01644
7.056	85.76544	12.213	8.375332	1.189651	0.459514	0.06527	-0.0199
8.064	85.3652	12.18447	9.334227	1.301378	0.514525	0.071735	-0.02338

9.072	86.1125	12.23768	10.6192	1.468966	0.580276	0.08027	-0.02689
10.08	86.73573	12.28189	11.57342	1.624199	0.627874	0.088115	-0.02912
11.088	86.96923	12.29841	12.46893	1.797185	0.674641	0.097238	-0.03066
12.096	88.21637	12.38627	4.607501	3.618572	0.245768	0.193018	0.041327
13.104	88.87275	12.43227	4.757131	3.814137	0.251875	0.201947	0.038963
14.112	89.28541	12.4611	5.510728	4.132414	0.290427	0.217787	0.040952
15.12	89.42514	12.47085	5.883189	4.332273	0.309572	0.227963	0.039395
16.128	89.83037	12.49907	6.142168	4.521616	0.321742	0.236853	0.037684
17.136	90.49768	12.54541	6.405947	4.796426	0.333085	0.249396	0.038973
18.144	89.58997	12.48233	6.513231	4.961217	0.342094	0.260578	0.038155
19.152	90.5167	12.54673	6.704954	5.222349	0.348559	0.271485	0.038238
20.16	91.94884	12.64559	7.237618	5.546062	0.370389	0.283823	0.040702

Table VIII. Derived data of wing at high speed without plate.

Alpha deg	DynPressure Pa	TrueAirspeed m/s	Lift N	Drag N	CL	CD	CM
0	153.4788	16.33768	0.042854	1.098473	0.001314	0.033678	0.004435
1.008	151.6627	16.24073	1.756582	1.157733	0.0545	0.03592	0.000932
2.016	152.7253	16.29752	3.776742	1.254812	0.116363	0.038661	-0.00295
3.024	154.133	16.37246	5.621519	1.373076	0.17162	0.041919	-0.00607
4.032	151.8126	16.24875	7.508649	1.507528	0.232736	0.046727	-0.00934
5.04	153.0735	16.31609	9.426876	1.65919	0.289786	0.051004	-0.01258
6.048	153.2503	16.32551	11.33945	1.841614	0.348177	0.056547	-0.01521
7.056	152.2842	16.27397	13.27814	2.056643	0.41029	0.06355	-0.01797
8.064	153.1626	16.32084	14.98118	2.263039	0.460259	0.069526	-0.0202
9.072	152.4652	16.28364	16.83997	2.497924	0.519732	0.077093	-0.02268
10.08	153.367	16.33172	18.76423	2.7901	0.575715	0.085605	-0.02472
11.088	153.4268	16.33491	20.42203	3.070745	0.626335	0.094178	-0.02669
12.096	152.5891	16.29026	22.02375	3.343791	0.679167	0.103116	-0.02753
13.104	153.3612	16.33142	24.05164	3.709985	0.737969	0.113832	-0.02855
14.112	152.8875	16.30618	25.76921	4.029971	0.793118	0.124033	-0.02933
15.12	154.5569	16.39496	27.40079	4.380694	0.834226	0.133372	-0.03003
16.128	154.2516	16.37876	29.26268	4.746111	0.892675	0.144783	-0.03133
17.136	155.2045	16.42927	30.58546	5.123616	0.927299	0.155339	-0.03119
18.144	153.2526	16.32563	31.43509	5.517936	0.965196	0.169425	-0.02899
19.152	159.1475	16.63665	12.84239	9.445704	0.379713	0.279282	0.047451
20.16	159.28	16.64358	14.14578	10.17375	0.417902	0.300558	0.051649
21.168	160.4404	16.7041	14.29244	10.55099	0.419181	0.309449	0.050418

22.176	159.9218	16.67707	15.02677	11.13138	0.442147	0.327529	0.053629
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Table IX. Derived data of wing at low speed without plate.

Alpha deg	DynPressure Pa	TrueAirspeed m/s	Lift N	Drag N	CL	CD	CM
0	85.56961	12.19905	0.047269	0.60471	0.002599	0.033253	0.004287
1.008	85.05163	12.16207	0.95289	0.632747	0.052719	0.035007	0.000742
2.016	85.82064	12.21693	2.078129	0.698783	0.113944	0.038314	-0.00308
3.024	86.10018	12.23681	3.130203	0.776539	0.171071	0.042439	-0.0061
4.032	85.39808	12.18681	4.208008	0.85355	0.231866	0.047032	-0.00906
5.04	86.75185	12.28303	5.248159	0.95072	0.284667	0.051568	-0.01245
6.048	85.75479	12.21224	6.289797	1.061672	0.345134	0.058256	-0.01517
7.056	85.19286	12.17216	7.296674	1.179668	0.403024	0.065158	-0.01844
8.064	85.80174	12.21558	8.276673	1.314373	0.453909	0.072083	-0.02116
9.072	86.02584	12.23152	9.249378	1.462742	0.505933	0.080011	-0.02421
10.08	85.96046	12.22687	10.20591	1.633623	0.558679	0.089426	-0.02636
11.088	85.43165	12.18921	10.99707	1.80978	0.605713	0.099682	-0.02777
12.096	88.4592	12.40331	5.124344	3.673809	0.272586	0.195426	0.048613
13.104	88.84062	12.43002	5.795057	4.002507	0.306941	0.211997	0.052553
14.112	89.31679	12.46329	5.7693	4.16517	0.303948	0.219436	0.047945
15.12	89.37072	12.46705	6.111428	4.376232	0.321778	0.230417	0.047326
16.128	89.66037	12.48724	6.701813	4.697135	0.351723	0.246514	0.04908
17.136	89.76261	12.49435	6.819486	4.906428	0.357491	0.257205	0.047889
18.144	89.86222	12.50128	7.163082	5.147703	0.375087	0.269554	0.0493
19.152	91.34033	12.60368	7.411947	5.446687	0.381838	0.280594	0.047887
20.16	91.00192	12.58031	7.645216	5.691218	0.395319	0.294282	0.047961

Interpreting the Data

By looking at Figure 1, it is clear that both the presence of the end plate as well as the air flow velocity have an impact on the lift coefficient of the wing. In all four configurations, the lift coefficient seems to have an increasing linear relationship with the angle of attack until the critical stalling angle. The wings experiencing high speed flow are able to experience a much greater angle of attack compared to those experiencing low speed flow. Additionally, the average

lift coefficient of wings with the end plates was notably higher than their plate-less counterparts at the same speed. Once the wing stalled, it is important to note that all four configurations experienced a dramatic decrease in lift coefficient.

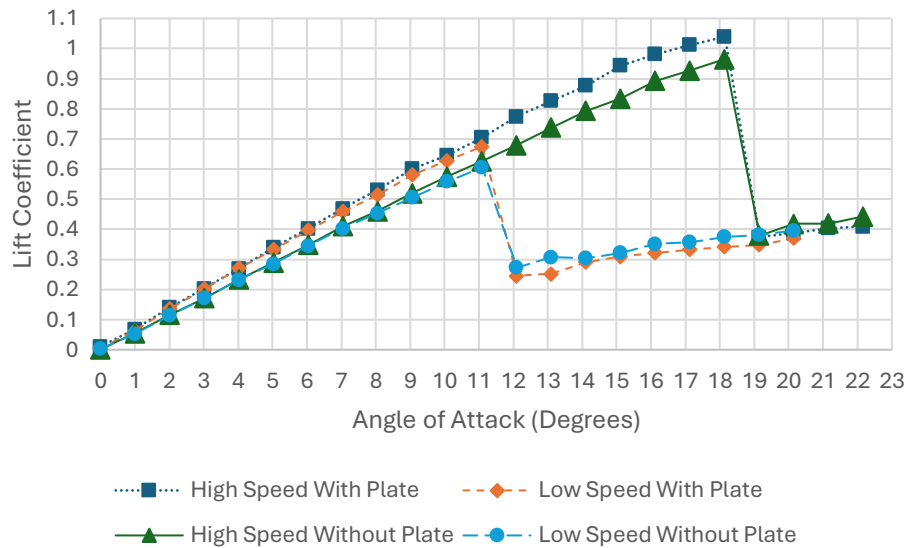


Figure 1. Lift coefficient vs. angle of attack for both wind speeds, with and without end plates.

By looking at Figure 2, it can be seen that the presence of an end plate makes a much smaller difference on the drag coefficient compared to the lift coefficient. With this being said, the drag coefficient of the wing with the plate is slightly lower than the drag coefficient of the wing without the plate, especially in the low-speed configurations. The wind speed, on the other hand, continues to play a large role in determining the drag coefficient of a wing. It is important to note that all four configurations had an increasing, linear relationship between the drag coefficient and the angle of attack, until the wing stalled. When stalled, the drag coefficient would spike dramatically. The wings experiencing a higher flow rate were able to maintain a lower drag coefficient for longer when compared to the wings experiencing low air flow rate.

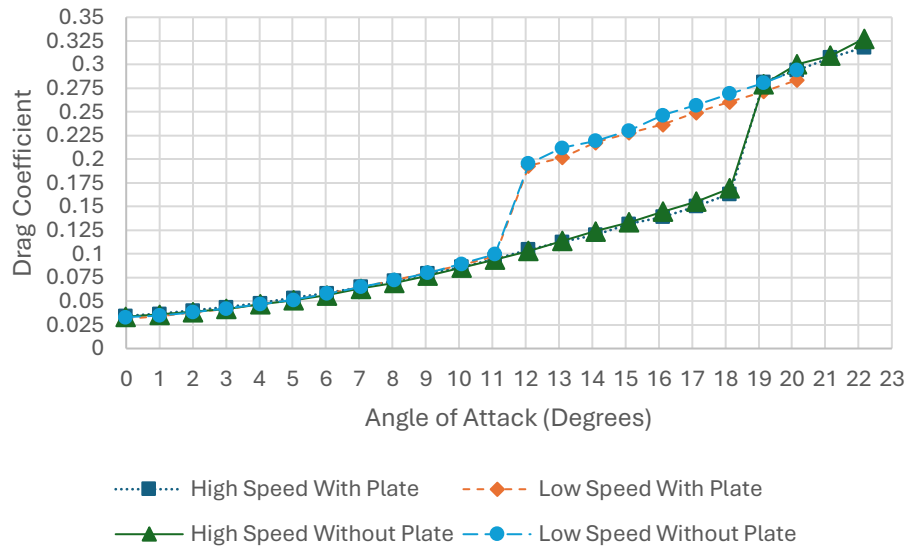


Figure 2. Drag coefficient vs. angle of attack for both wind speeds, with and without end plates.

Looking at Figure 3, both the presence of the end plate and the different air flow velocities seem to have a major impact on the aerodynamic performance of the wing. The best performing configuration was the wing containing an end plate and experiencing high wind speeds. An ideal aerodynamic wing would try to maximize its lift coefficient while minimizing its drag coefficient. The end plate & high wind speed configuration is by far the most efficient, only really challenged by the no end plate & high wind speed configuration. Based on this figure, it is clear that wind speed is the most significant determining factor for aerodynamic performance because it enables a wing to experience higher angles of attack and maximize its lift coefficient. Additionally, the presence of an end plate gives a slight but notable edge in efficiency. It is important to note that these aerodynamic differences occur before the wing stalls, and the wing efficiency dramatically drops afterwards.

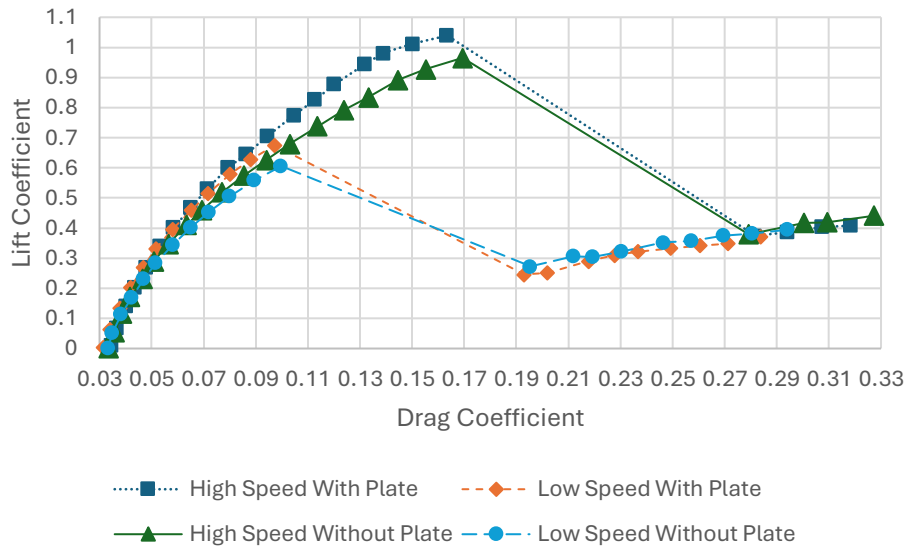


Figure 3. Lift coefficient vs. drag coefficient for both wind speeds, with and without end plates.

Looking at Figure 4, an interesting relationship occurs between the moment coefficient and the angle of attack. Before stalling, the moment coefficient consistently decreases. This means that the aerodynamic center of the wing is actually moving a little behind the quarter-chord of the wing. This makes sense because the increase in lift generated by the increase in angle of attack should result in a downwards/negative moment as the wing naturally stabilizes. Once the wing stalls, the moment coefficient suddenly becomes positive, and the aerodynamic center of the wing actually moves in front of the quarter-chord. This also makes logical sense that the wing loses a considerable amount of lift, creating an upwards/positive moment. Additionally, it is important to note that wind speed plays a large role in determining the moment coefficient of a wing because those that experience higher wind speeds can experience a greater angle of attack before stalling.

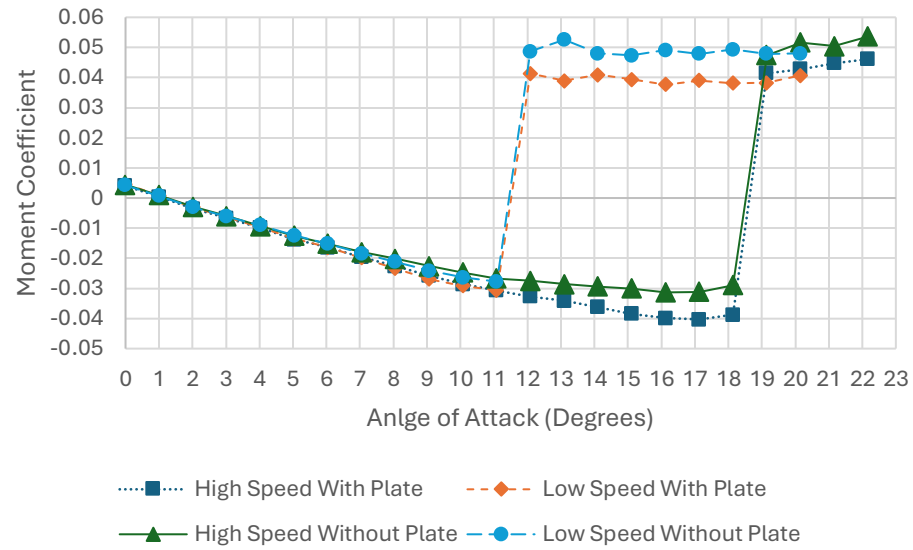


Figure 4. Moment coefficient vs angle of attack for both wind speeds, with & without end plates.

Summary and Conclusions

In conclusion, this experiment was centered around determining the aerodynamic forces like lift and drag as well as the non-dimensional aerodynamic coefficients of a symmetric wing in a low-turbulence, subsonic wind tunnel. Additionally, this lab aimed to quantitatively evaluate the impact of end plates and different airstream velocities on the performance of a wing. Equipped with the combination of a six-component load cell and a Pitot-static probe system, this experiment was able to precisely measure lift, drag, and pitching moment over thousands of samples across angles of attack from 0° to 22° . The model was tested in four main configurations, with and without a perpendicular end plate as well as at air flow velocities of either 12.5 m/s and 16.5 m/s.

All in all, the resulting aerodynamic coefficients agreed well with established aerodynamic findings like the thin airfoil theory while also accurately portraying the real-world impact of varying flow speeds and end plates. The end plate was responsible for physically blocking flow around the span of the wing, reducing vortex formation, and effectively increasing the wing's aerodynamic performance. The inclusion of an end plate produced a steeper slope on the lift vs. drag coefficient curve as well as a clear reduction in induced drag at the highest lift coefficients.

The derived data also showed the considerable impact of freestream velocity. Air speed was by far the main influence on the wing's stall behavior. The higher wind speed delayed flow separation and raised the critical stall angle from approximately 12° to 19° . The analysis of the coefficient of the pitching moment further confirmed the findings of this report. It can be seen that the aerodynamic center sat slightly aft of the quarter-chord mounting point, indicated by a steady negative moment slope before stall that reversed sharply positive once the flow separated.

Overall, this experiment illustrated the importance of both airflow velocity and end plates in the performance of a wing in a controlled environment like a wind tunnel. It showed that end plates are an effective way to improve wing efficiency at higher angles of attack, while also highlighting how faster air flow can dramatically increase the angle of stall as well as the maximum lift coefficient.

Appendix: Error Analysis

Random Errors

The random errors found in this experiment were best minimized through repetition. At each angle throughout the experiment, the stepper would record data for 3 seconds and the LabView VI would log 1000 samples per second of each of the six load cell measurements. As mentioned earlier, a MATLAB script was created to properly organize the thousands of data points as well as calculate the mean and variance of the Baratron voltage as well as each component of force and torque as seen in Table I. To find the standard error of the mean (1 SEM), simply take the square root of the variance before dividing by the square root of the number of samples (3000) as seen in the following equations:

$$\text{Variance} = \sigma^2$$

$$U_{rand} = \frac{\sigma}{\sqrt{N}}$$

To get a 95% Confidence Interval, simply make sure to multiply the random error by 2 (or 1.96) to account for two standard errors of the mean (2 SEMs). This relationship can be seen in the following equation:

$$U_{rand_95} = 2 * \left(\frac{\sigma}{\sqrt{N}} \right)$$

Finally, to calculate all the random errors found this experiment, simply apply the equation above to the x-,y-,z- components of the force and the x-,y-,z- components of the torque, and the Baratron voltage data for the four different configurations (High vs Low Speed, With vs. Without Plate). The accumulation of all random error in this experiment can be found in the tables below:

*Units: Force = N, T = N-m, Bar = V

Table X. Random error calculations for high speed with plate

Alpha_deg	FxRan	FyRan	FzRan	TxRan	TyRan	TzRan	BarRan
0	0.009396	0.005404	0.004766	0.000543	0.002022	0.000422	0.00037
1.008	0.008884	0.003923	0.004515	0.000665	0.001662	0.00036	0.000231
2.016	0.008583	0.003607	0.004413	0.000882	0.002367	0.000458	0.000157
3.024	0.008006	0.003351	0.004538	0.000534	0.0021	0.00049	0.000345
4.032	0.008707	0.003834	0.004938	0.000635	0.002188	0.000448	0.000211
5.04	0.006554	0.003115	0.004785	0.000687	0.001815	0.000456	0.000174
6.048	0.009869	0.003693	0.004268	0.001047	0.0033	0.000527	0.000265
7.056	0.007226	0.004302	0.004792	0.001434	0.002107	0.000571	0.000196
8.064	0.007613	0.003142	0.004312	0.000785	0.002149	0.000548	0.000513
9.072	0.011146	0.00378	0.00514	0.001345	0.004551	0.000734	0.000333
10.08	0.011096	0.00384	0.00445	0.001158	0.004428	0.00072	0.000175
11.088	0.01084	0.003707	0.004911	0.001225	0.004114	0.000791	0.000418
12.096	0.008449	0.005075	0.005023	0.001768	0.002918	0.000544	0.000186
13.104	0.018126	0.005519	0.004555	0.002296	0.007883	0.001188	0.00026
14.112	0.013744	0.007725	0.004795	0.003411	0.005811	0.000955	0.000202
15.12	0.015901	0.011382	0.005566	0.004999	0.006892	0.001048	0.000194
16.128	0.009333	0.007879	0.006311	0.003339	0.003522	0.000774	0.000169
17.136	0.021801	0.007344	0.120742	0.003797	0.00611	0.001408	0.00019

18.144	0.016962	0.006144	0.139483	0.002288	0.002464	0.001063	0.000225
19.152	0.034354	0.007381	0.009324	0.0034	0.014991	0.002628	0.000162
20.16	0.036919	0.007261	0.010138	0.003143	0.016819	0.002538	0.000288
21.168	0.021796	0.014764	0.008129	0.006693	0.009432	0.00176	0.000192
22.176	0.040659	0.009161	0.009045	0.003818	0.018258	0.002908	0.000311

Table XI. Random error calculations for low speed with plate

Alpha_deg	FxRan	FyRan	FzRan	TxRan	TyRan	TzRan	BarRan
0	0.006589	0.002465	0.003185	0.000389	0.002129	0.000405	2.09E-04
1.008	0.004735	0.002527	0.003018	0.000493	0.001004	0.000277	1.63E-04
2.016	0.004798	0.002345	0.003084	0.00047	0.001218	0.000315	2.37E-04
3.024	0.004938	0.002421	0.003026	0.000461	0.001392	0.000395	1.56E-04
4.032	0.005894	0.002335	0.003062	0.000414	0.001852	0.000384	1.46E-04
5.04	0.005427	0.002268	0.003014	0.000683	0.001813	0.000351	1.59E-04
6.048	0.004864	0.002076	0.003067	0.000408	0.00151	0.000408	1.59E-04
7.056	0.006506	0.002135	0.003269	0.000343	0.002368	0.000465	1.93E-04
8.064	0.005387	0.00254	0.003223	0.00083	0.00167	0.000376	1.63E-04
9.072	0.006062	0.003442	0.003154	0.001346	0.002264	0.000443	1.96E-04
10.08	0.005256	0.00237	0.0031	0.000697	0.001619	0.000489	1.78E-04
11.088	0.006318	0.002218	0.003292	0.000479	0.002027	0.000472	1.88E-04
12.096	0.051873	0.009946	0.016919	0.00416	0.02336	0.003734	1.41E-04
13.104	0.02669	0.009004	0.015783	0.003294	0.012099	0.001911	1.64E-04
14.112	0.024202	0.008596	0.012988	0.003345	0.010711	0.00166	1.38E-04
15.12	0.032191	0.005597	0.009487	0.002	0.014039	0.002289	2.21E-04
16.128	0.024712	0.00913	0.009455	0.003904	0.01083	0.001872	1.37E-04
17.136	0.026419	0.006402	0.008884	0.00271	0.01183	0.001893	1.83E-04
18.144	0.034486	0.008651	0.008789	0.004013	0.015515	0.002228	1.51E-04
19.152	0.017106	0.006206	0.006578	0.002468	0.007015	0.001169	1.41E-04
20.16	0.032744	0.006261	0.006169	0.002752	0.014782	0.002382	1.57E-04

Table XII. Random error calculations for high speed without plate.

Alpha_deg	FxRan	FyRan	FzRan	TxRan	TyRan	TzRan	BarRan
0	0.007967	0.004151	0.004977	0.001044	0.001196	0.000467	2.21E-04
1.008	0.008179	0.004616	0.005386	0.000901	0.001457	0.000494	2.78E-04
2.016	0.008131	0.003789	0.00455	0.000766	0.001807	0.000409	2.96E-04
3.024	0.00767	0.004209	0.005111	0.000783	0.001518	0.000433	1.98E-04
4.032	0.007658	0.004127	0.006489	0.001249	0.002004	0.000501	1.71E-04

5.04	0.006655	0.00319	0.005269	0.000837	0.001405	0.000393	2.31E-04
6.048	0.010563	0.003822	0.005861	0.001076	0.003744	0.00065	2.41E-04
7.056	0.007029	0.002632	0.004953	0.000543	0.001666	0.000525	2.21E-04
8.064	0.009855	0.00539	0.004496	0.00212	0.003433	0.000644	2.80E-04
9.072	0.009772	0.003129	0.004771	0.000764	0.003271	0.000643	2.06E-04
10.08	0.011148	0.00312	0.004905	0.000583	0.003604	0.000764	2.70E-04
11.088	0.011852	0.003087	0.00484	0.000781	0.004129	0.000799	1.96E-04
12.096	0.011582	0.003586	0.005504	0.001217	0.004168	0.000737	3.95E-04
13.104	0.011562	0.00354	0.005828	0.00089	0.004113	0.000793	3.27E-04
14.112	0.009421	0.003445	0.005622	0.000862	0.003025	0.000641	2.49E-04
15.12	0.014513	0.003861	0.006288	0.001129	0.005356	0.000923	4.87E-04
16.128	0.012831	0.003881	0.004884	0.001189	0.004919	0.000862	3.26E-04
17.136	0.012009	0.003615	0.005609	0.001255	0.004585	0.000864	2.15E-04
18.144	0.019174	0.006311	0.005685	0.002414	0.007596	0.001351	2.31E-04
19.152	0.048581	0.021027	0.008468	0.008774	0.019649	0.0037	3.10E-04
20.16	0.045039	0.008616	0.007237	0.003291	0.019629	0.003557	3.11E-04
21.168	0.038173	0.016938	0.006456	0.00717	0.015702	0.002509	2.35E-04
22.176	0.0378	0.01661	0.006145	0.006907	0.014688	0.002288	3.35E-04

Table XIII. Random error calculations for low speed without plate.

Alpha_deg	FxRan	FyRan	FzRan	TxRan	TyRan	TzRan	BarRan
0	0.004985	0.002856	0.003147	0.000514	0.001038	0.000434	4.78E-04
1.008	0.004923	0.002423	0.003361	0.000543	0.00108	0.000304	5.03E-04
2.016	0.004481	0.002466	0.003104	0.000656	0.000998	0.000339	4.51E-04
3.024	0.004593	0.003205	0.003106	0.000937	0.001054	0.000328	5.05E-04
4.032	0.00472	0.002319	0.002973	0.000574	0.001465	0.000346	4.10E-04
5.04	0.00425	0.002253	0.00311	0.000658	0.001098	0.000299	4.96E-04
6.048	0.004883	0.002276	0.003073	0.000557	0.001644	0.000467	4.46E-04
7.056	0.005345	0.002272	0.003219	0.00071	0.001702	0.000427	4.96E-04
8.064	0.005407	0.002471	0.003445	0.000752	0.001502	0.000488	5.41E-04
9.072	0.005383	0.003123	0.003234	0.001125	0.001526	0.000435	4.99E-04
10.08	0.006059	0.002558	0.003294	0.0006	0.002018	0.000518	4.31E-04
11.088	0.010311	0.002993	0.003394	0.001017	0.003932	0.00071	4.44E-04
12.096	0.026647	0.007976	0.004874	0.0034	0.011064	0.001779	4.79E-04
13.104	0.022316	0.004423	0.004573	0.001756	0.009166	0.001979	4.70E-04
14.112	0.024492	0.004044	0.004632	0.001646	0.010341	0.002104	4.26E-04
15.12	0.019884	0.004679	0.004366	0.001897	0.008588	0.001676	4.93E-04
16.128	0.02467	0.009072	0.004851	0.00378	0.010217	0.001935	5.24E-04
17.136	0.032558	0.006954	0.005579	0.002925	0.01364	0.002158	4.74E-04
18.144	0.029981	0.009473	0.005269	0.004004	0.012611	0.002171	4.64E-04

19.152	0.022999	0.003619	0.004686	0.001523	0.009296	0.001675	4.38E-04
20.16	0.021128	0.007423	0.004276	0.003052	0.008528	0.001469	4.70E-04

Systematic Errors

The systematic/bias errors found in this experiment are unavoidable and cannot be minimized through repetition. These errors are very important because they consistently alter each of the measurements taken in the same way. Systematic errors are predicted and caused by the physical limitations and calibrations of the laboratory equipment. The sensors used contain systematic uncertainties that were provided prior to the lab and can be found below:

- Baratron Voltage: ± 0.002 V
- Load/Forces: ± 0.05 N
- Torque/Moments: ± 0.001 N-m

Moving forward, it is extremely important to note that systematic errors must be accounted for with the total error calculations for every instance of random error. This means that random Baratron voltage error must account for ± 0.002 V, the random force error must account for ± 0.05 N, and the random torque error must account for ± 0.001 N-m.

Total Error

Raw Data Total Error

In order to calculate the total error found within this experiment, it is essential to account for all instances of random and systematic error. The first step revolves around calculating the total error from the raw data. This means accounting for the total error of 7 different aspects (3 components of force, 3 components of torque, 1 Baratron Voltage) of the model wing in the four

configurations (High vs Low Speed, With vs Without Plate). It is important to note that the total uncertainty created by the raw data measurements can be represented in the following equation:

$$U_{total} = \sqrt{U_{systematic}^2 + U_{rand}^2}$$

To calculate the total error of all the raw data points, simply apply this equation using the systematic error provided as well as the derived random error. To reinforce, random Baratron voltage error must account for +/- 0.002 V, the random force error must account for +/- 0.05 N, and the random torque error must account for +/- 0.001 N-m. These total error results for the raw measurements can be found in the following tables:

*Units: Force = N, T = N-m, Bar = V

Table XIV. Total error for raw data points for high speed with plate.

Alpha deg	FxTot	FyTot	FzTot	TxTot	TyTot	TzTot	BarTot
0	0.050875	0.050291	0.050227	0.001138	2.26E-03	0.001085	2.03E-03
1.008	0.050783	0.050154	0.050203	0.001201	1.94E-03	0.001063	2.01E-03
2.016	0.050731	0.05013	0.050194	0.001333	2.57E-03	0.0011	2.01E-03
3.024	0.050637	0.050112	0.050206	0.001134	2.33E-03	0.001114	2.03E-03
4.032	0.050752	0.050147	0.050243	0.001185	2.41E-03	0.001096	2.01E-03
5.04	0.050428	0.050097	0.050228	0.001213	2.07E-03	0.001099	2.01E-03
6.048	0.050965	0.050136	0.050182	0.001448	3.45E-03	0.001131	2.02E-03
7.056	0.05052	0.050185	0.050229	0.001748	2.33E-03	0.001151	2.01E-03
8.064	0.050576	0.050099	0.050186	0.001271	2.37E-03	0.001141	2.06E-03
9.072	0.051227	0.050143	0.050263	0.001676	4.66E-03	0.001241	2.03E-03
10.08	0.051217	0.050147	0.050198	0.00153	4.54E-03	0.001232	2.01E-03
11.088	0.051161	0.050137	0.050241	0.001581	4.23E-03	0.001275	2.04E-03
12.096	0.050709	0.050257	0.050252	0.002031	3.08E-03	0.001139	2.01E-03
13.104	0.053184	0.050304	0.050207	0.002505	7.95E-03	0.001553	2.02E-03
14.112	0.051855	0.050593	0.050229	0.003555	5.90E-03	0.001383	2.01E-03
15.12	0.052467	0.051279	0.050309	0.005098	6.96E-03	0.001449	2.01E-03
16.128	0.050864	0.050617	0.050397	0.003486	3.66E-03	0.001264	2.01E-03
17.136	0.054546	0.050536	0.130685	0.003927	6.19E-03	0.001727	2.01E-03
18.144	0.052799	0.050376	0.148174	0.002497	2.66E-03	0.001459	2.01E-03
19.152	0.060665	0.050542	0.050862	0.003544	1.50E-02	0.002812	2.01E-03

20.16	0.062153	0.050524	0.051017	0.003298	1.68E-02	0.002728	2.02E-03
21.168	0.054544	0.052134	0.050656	0.006768	9.48E-03	0.002025	2.01E-03
22.176	0.064445	0.050832	0.050812	0.003947	1.83E-02	0.003075	2.02E-03

Table XV. Total error for raw data points for low speed with plate.

Alpha_deg	FxTot	FyTot	FzTot	TxTot	TyTot	TzTot	BarTot
0	0.050432	0.050061	0.050101	0.050442	0.050071	0.050111	0.002011
1.008	0.050224	0.050064	0.050091	0.050234	0.050074	0.050101	0.002007
2.016	0.05023	0.050055	0.050095	0.05024	0.050065	0.050105	0.002014
3.024	0.050243	0.050059	0.050092	0.050253	0.050069	0.050101	0.002006
4.032	0.050346	0.050054	0.050094	0.050356	0.050064	0.050104	0.002005
5.04	0.050294	0.050051	0.050091	0.050304	0.050061	0.050101	0.002006
6.048	0.050236	0.050043	0.050094	0.050246	0.050053	0.050104	0.002006
7.056	0.050421	0.050046	0.050107	0.050431	0.050056	0.050117	0.002009
8.064	0.050289	0.050064	0.050104	0.050299	0.050074	0.050114	0.002007
9.072	0.050366	0.050118	0.050099	0.050376	0.050128	0.050109	0.00201
10.08	0.050275	0.050056	0.050096	0.050285	0.050066	0.050106	0.002008
11.088	0.050398	0.050049	0.050108	0.050408	0.050059	0.050118	0.002009
12.096	0.072047	0.05098	0.052785	0.072054	0.050989	0.052795	0.002005
13.104	0.056678	0.050804	0.052432	0.056687	0.050814	0.052441	0.002007
14.112	0.055549	0.050734	0.051659	0.055558	0.050743	0.051669	0.002005
15.12	0.059466	0.050312	0.050892	0.059475	0.050322	0.050902	0.002012
16.128	0.055774	0.050827	0.050886	0.055783	0.050837	0.050896	0.002005
17.136	0.05655	0.050408	0.050783	0.056559	0.050418	0.050793	0.002008
18.144	0.06074	0.050743	0.050767	0.060748	0.050753	0.050777	0.002006
19.152	0.052845	0.050384	0.050431	0.052855	0.050394	0.050441	0.002005
20.16	0.059767	0.05039	0.050379	0.059776	0.0504	0.050389	0.002006

Table XVI. Total error for raw data points for high speed without plate.

Alpha_deg	FxTot	FyTot	FzTot	TxTot	TyTot	TzTot	BarTot
0	0.050631	0.050172	0.050247	0.001445	0.050182	0.050257	0.002012
1.008	0.050664	0.050213	0.050289	0.001346	0.050223	0.050299	0.002019
2.016	0.050657	0.050143	0.050207	0.00126	0.050153	0.050217	0.002022
3.024	0.050585	0.050177	0.050261	0.00127	0.050187	0.050271	0.00201
4.032	0.050583	0.05017	0.050419	0.0016	0.05018	0.050429	0.002007
5.04	0.050441	0.050102	0.050277	0.001304	0.050112	0.050287	0.002013
6.048	0.051104	0.050146	0.050342	0.001469	0.050156	0.050352	0.002014
7.056	0.050492	0.050069	0.050245	0.001138	0.050079	0.050255	0.002012

8.064	0.050962	0.05029	0.050202	0.002344	0.0503	0.050212	0.002019
9.072	0.050946	0.050098	0.050227	0.001259	0.050108	0.050237	0.002011
10.08	0.051228	0.050097	0.05024	0.001158	0.050107	0.05025	0.002018
11.088	0.051386	0.050095	0.050234	0.001269	0.050105	0.050244	0.00201
12.096	0.051324	0.050128	0.050302	0.001575	0.050138	0.050312	0.002039
13.104	0.051319	0.050125	0.050339	0.001339	0.050135	0.050348	0.002027
14.112	0.05088	0.050119	0.050315	0.00132	0.050128	0.050325	0.002015
15.12	0.052064	0.050149	0.050394	0.001508	0.050159	0.050404	0.002058
16.128	0.05162	0.05015	0.050238	0.001553	0.05016	0.050248	0.002026
17.136	0.051422	0.050131	0.050314	0.001605	0.05014	0.050324	0.002011
18.144	0.05355	0.050397	0.050322	0.002612	0.050407	0.050332	0.002013
19.152	0.069715	0.054242	0.050712	0.008831	0.054251	0.050722	0.002024
20.16	0.067294	0.050737	0.050521	0.00344	0.050747	0.050531	0.002024
21.168	0.062906	0.052791	0.050415	0.007239	0.052801	0.050425	0.002014
22.176	0.06268	0.052687	0.050376	0.006979	0.052696	0.050386	0.002028

Table XVII. Total error for raw data points for low speed without plate.

Alpha deg	FxTot	FyTot	FzTot	TxTot	TyTot	TzTot	BarTot
0	0.050248	0.050082	0.050099	0.001124	0.001441	0.00109	0.002056
1.008	0.050242	0.050059	0.050113	0.001138	0.001472	0.001045	0.002062
2.016	0.0502	0.050061	0.050096	0.001196	0.001413	0.001056	0.00205
3.024	0.050211	0.050103	0.050096	0.00137	0.001453	0.001052	0.002063
4.032	0.050222	0.050054	0.050088	0.001153	0.001774	0.001058	0.002042
5.04	0.05018	0.050051	0.050097	0.001197	0.001485	0.001044	0.002061
6.048	0.050238	0.050052	0.050094	0.001145	0.001924	0.001104	0.002049
7.056	0.050285	0.050052	0.050103	0.001226	0.001974	0.001087	0.002061
8.064	0.050292	0.050061	0.050119	0.001251	0.001804	0.001113	0.002072
9.072	0.050289	0.050097	0.050104	0.001505	0.001824	0.001091	0.002061
10.08	0.050366	0.050065	0.050108	0.001166	0.002252	0.001126	0.002046
11.088	0.051052	0.05009	0.050115	0.001426	0.004057	0.001226	0.002049
12.096	0.056657	0.050632	0.050237	0.003544	0.01111	0.002041	0.002057
13.104	0.054754	0.050195	0.050209	0.002021	0.009221	0.002218	0.002054
14.112	0.055677	0.050163	0.050214	0.001926	0.01039	0.00233	0.002045
15.12	0.053809	0.050218	0.05019	0.002145	0.008646	0.001952	0.00206
16.128	0.055755	0.050816	0.050235	0.00391	0.010266	0.002178	0.002068
17.136	0.059666	0.050481	0.05031	0.003091	0.013677	0.002379	0.002055
18.144	0.0583	0.050889	0.050277	0.004127	0.01265	0.00239	0.002053
19.152	0.055036	0.050131	0.050219	0.001822	0.00935	0.001951	0.002047
20.16	0.054281	0.050548	0.050183	0.003211	0.008586	0.001777	0.002054

Reduced Data Total Error

The second step for calculating total error is to calculate the total error for the reduced data. There were three main types of reduced data error that needed to be calculated.

First, the total force error for both lift and drag need to be calculated and considered. These calculations are made possible by properly integrating the total raw data error for the x- and y- components of force in each of the four configurations. The process of calculating the total error that propagates through lift and drag mirrors the lift and drag force calculation seen in the Results & Discussion section. The key difference between calculating lift/drag force and lift/drag total error is that the x- and y- force components are replaced by their respective total errors. This change can best be seen in the following two expressions below:

$$U_L = \sqrt{(U_{F_X} \cos \alpha)^2 + (U_{F_Y} \sin \alpha)^2}$$

$$U_D = \sqrt{(U_{F_X} \sin \alpha)^2 + (U_{F_Y} \cos \alpha)^2}$$

Second, it is important to clarify the total pitching moment error. Given that the wing is vertical and that the moment is directly around the z-axis, the total error of the pitching moment can simply be identified as the total error of the z-component of the torque. This relationship is illustrated in the equation below:

$$U_M = U_{T_Z}$$

Additionally, the total error of the dynamic pressure must also be calculated. Given that the Baratron has a known sensitivity of 1.016 mmHg or 135.455 Pa, the propagation of total error involving the dynamic pressure can be represented in the equation below:

$$U_q = U_{Voltage} \times 135.455$$

Third, the total error of the non-dimensional aerodynamic coefficients must be considered. All the following formulas involve the integration of two different sources of error. Each equation must consider their respective total error caused by either lift, drag, or moment. Additionally, each equation must consider the propagation of the dynamic pressure total error as well. All in all, the three equations for calculating the total error of non-dimensional coefficients mirror those mentioned in the Results & Discussion section, with a few tweaks to consider the two different sources of error propagation.

$$U_{C_L} = \frac{1}{qS} \sqrt{U_L^2 + (C_L \cdot U_q \cdot S)^2}$$

$$U_{C_D} = \frac{1}{qS} \sqrt{U_D^2 + (C_D \cdot U_q \cdot S)^2}$$

$$U_{C_M} = \frac{1}{qSc} \sqrt{U_M^2 + (C_M \cdot U_q \cdot S \cdot c)^2}$$

Putting this all together in the form of a MATLAB script, the following tables were able to be generated. In these tables contain all the information the total error that could be derived from the reduced data.

*Units: Lift/Drag = N, Dynamic Pressure = Pa

Table XVIII. Reduced data total error for high speed with plate.

Alpha deg	Lift	Drag	DynPress	CL	CD	CM
0	0.050875	5.03E-02	2.76E-01	1.56E-03	1.55E-03	1.08E-04
1.008	0.050334	5.06E-02	2.73E-01	1.55E-03	1.56E-03	1.06E-04
2.016	0.050242	5.06E-02	2.72E-01	1.56E-03	1.55E-03	1.09E-04

3.024	0.05063	5.01E-02	2.75E-01	1.60E-03	1.54E-03	1.11E-04
4.032	0.050387	5.05E-02	2.72E-01	1.61E-03	1.55E-03	1.09E-04
5.04	0.050131	5.04E-02	2.72E-01	1.66E-03	1.56E-03	1.12E-04
6.048	0.05092	5.02E-02	2.73E-01	1.73E-03	1.56E-03	1.17E-04
7.056	0.050357	5.03E-02	2.72E-01	1.76E-03	1.55E-03	1.20E-04
8.064	0.050119	5.06E-02	2.80E-01	1.83E-03	1.57E-03	1.21E-04
9.072	0.051099	5.03E-02	2.75E-01	1.91E-03	1.56E-03	1.32E-04
10.08	0.050822	5.05E-02	2.72E-01	1.94E-03	1.56E-03	1.32E-04
11.088	0.050146	5.12E-02	2.77E-01	1.99E-03	1.58E-03	1.38E-04
12.096	0.050616	5.04E-02	2.72E-01	2.09E-03	1.57E-03	1.28E-04
13.104	0.052444	5.11E-02	2.73E-01	2.18E-03	1.58E-03	1.66E-04
14.112	0.050594	5.19E-02	2.72E-01	2.19E-03	1.60E-03	1.50E-04
15.12	0.052105	5.16E-02	2.72E-01	2.34E-03	1.62E-03	1.61E-04
16.128	0.050823	5.07E-02	2.72E-01	2.33E-03	1.57E-03	1.44E-04
17.136	0.050621	5.45E-02	2.72E-01	2.34E-03	1.67E-03	1.83E-04
18.144	0.051794	5.14E-02	2.73E-01	2.41E-03	1.59E-03	1.58E-04
19.152	0.059836	5.15E-02	2.72E-01	1.89E-03	1.60E-03	2.78E-04
20.16	0.051377	6.15E-02	2.74E-01	1.65E-03	1.87E-03	2.68E-04
21.168	0.053262	5.34E-02	2.72E-01	1.70E-03	1.64E-03	2.05E-04
22.176	0.064032	5.14E-02	2.74E-01	1.99E-03	1.59E-03	3.00E-04

Table XIX. Reduced data total error for low speed with plate.

Alpha_deg	Lift	Drag	DynPress	CL	CD	CM
0	0.050432	0.050061	0.272391	0.002735	0.002717	0.00877
1.008	0.050109	0.050178	0.271806	0.002755	0.002754	0.008866
2.016	0.050087	0.050197	0.272808	0.002781	0.002756	0.008869
3.024	0.050241	0.050061	0.271728	0.002847	0.002767	0.008925
4.032	0.05017	0.050231	0.271629	0.002884	0.002761	0.008874
5.04	0.050077	0.050269	0.271768	0.002949	0.002769	0.008889
6.048	0.050226	0.050054	0.271764	0.003033	0.002756	0.008884
7.056	0.050239	0.050229	0.272167	0.003118	0.002764	0.008874
8.064	0.050074	0.05028	0.271806	0.00321	0.002781	0.008915
9.072	0.050337	0.050148	0.272213	0.003306	0.002752	0.008837
10.08	0.050194	0.050138	0.271975	0.00336	0.002734	0.008773
11.088	0.050052	0.050395	0.272108	0.003434	0.002744	0.008752
12.096	0.068253	0.055958	0.271582	0.003719	0.003044	0.009089
13.104	0.055198	0.052408	0.27182	0.003023	0.002843	0.008962
14.112	0.050737	0.055547	0.271553	0.002816	0.003002	0.008789
15.12	0.056807	0.053296	0.272553	0.003135	0.002889	0.008645
16.128	0.054982	0.051682	0.271549	0.00304	0.002801	0.008605
17.136	0.05054	0.056433	0.272047	0.002812	0.003029	0.008524

18.144	0.056751	0.055168	0.271683	0.003156	0.003004	0.008608
19.152	0.052632	0.050607	0.271579	0.002929	0.002754	0.008463
20.16	0.051065	0.059192	0.271743	0.002833	0.003143	0.008323

Table XX. Reduced data total error for high speed without plate.

Alpha deg	Lift	Drag	DynPress	CL	CD	CM
0	0.050631	0.050172	0.272562	0.001552	0.00154	0.004973
1.008	0.050342	0.050536	0.273519	0.001565	0.001569	0.005037
2.016	0.050239	0.050562	0.273854	0.001562	0.001559	0.004993
3.024	0.050579	0.050183	0.27224	0.001574	0.001534	0.004953
4.032	0.050334	0.05042	0.271904	0.001615	0.001565	0.005045
5.04	0.050137	0.050406	0.272715	0.001626	0.001552	0.004989
6.048	0.051052	0.050198	0.272874	0.001686	0.001545	0.00499
7.056	0.050286	0.050276	0.272554	0.001719	0.001558	0.005012
8.064	0.050319	0.050933	0.273547	0.001751	0.00157	0.004979
9.072	0.050845	0.0502	0.272343	0.001823	0.001556	0.005004
10.08	0.050811	0.05052	0.273375	0.001866	0.001558	0.004976
11.088	0.050106	0.051375	0.272211	0.001897	0.001585	0.004973
12.096	0.051081	0.050376	0.276133	0.001998	0.001565	0.005007
13.104	0.051009	0.050441	0.274514	0.002048	0.001561	0.004986
14.112	0.050119	0.050879	0.273008	0.002094	0.001582	0.004999
15.12	0.051482	0.050746	0.278821	0.002173	0.001564	0.004953
16.128	0.051379	0.050398	0.274485	0.002232	0.001559	0.004947
17.136	0.050157	0.051396	0.272467	0.002228	0.001582	0.004924
18.144	0.052247	0.051746	0.272703	0.00235	0.001617	0.004988
19.152	0.068483	0.055788	0.274151	0.002128	0.001718	0.004841
20.16	0.051997	0.066325	0.274174	0.001696	0.002027	0.004819
21.168	0.057688	0.058448	0.272774	0.001836	0.001793	0.004774
22.176	0.062369	0.053055	0.274684	0.001986	0.001659	0.004786

Table XXI. Reduced data total error for low speed without plate.

Alpha deg	Lift	Drag	DynPress	CL	CD	CM
0	0.050248	0.050082	0.278552	0.002763	0.002756	0.000194
1.008	0.050111	0.05019	0.279344	0.002778	0.002779	0.000187
2.016	0.050087	0.050175	0.277721	0.002771	0.002754	0.000187
3.024	0.050209	0.050104	0.279406	0.0028	0.002742	0.000187
4.032	0.050121	0.050156	0.276531	0.002862	0.002768	0.00019

5.04	0.050064	0.050167	0.279115	0.002866	0.002726	0.000187
6.048	0.050228	0.050062	0.277555	0.002974	0.002754	0.000202
7.056	0.050171	0.050165	0.279107	0.00307	0.002779	0.000203
8.064	0.050071	0.050282	0.28065	0.003122	0.002768	0.000209
9.072	0.050266	0.05012	0.27922	0.003203	0.002754	0.000208
10.08	0.050254	0.050177	0.277123	0.003288	0.002762	0.000216
11.088	0.050098	0.051044	0.277513	0.003389	0.00283	0.000236
12.096	0.055473	0.051927	0.278571	0.003073	0.00283	0.000382
13.104	0.053596	0.05143	0.278276	0.002997	0.002804	0.000413
14.112	0.050167	0.055673	0.276973	0.002806	0.003011	0.000423
15.12	0.05273	0.05135	0.279016	0.002953	0.002798	0.000363
16.128	0.054964	0.05167	0.280053	0.003087	0.002819	0.000399
17.136	0.050684	0.059494	0.278426	0.002879	0.003219	0.000429
18.144	0.055305	0.054129	0.278105	0.00312	0.002955	0.000432
19.152	0.054618	0.050585	0.277331	0.003043	0.002742	0.000355
20.16	0.050804	0.054041	0.278279	0.002892	0.002936	0.000331

Most Significant Source of Error

Systematic error was consistently found to be the most significant source of error throughout this experiment. Given that the set up for this experiment allowed for thousands of samples collected for each angle of attack, the random error found in the raw data was mostly minimized. By looking at the charts provided in this section, it should be clear that the random errors were magnitudes of order smaller than the unavoidable systematic error. More specifically, the systematic error of the Load Cell (0.005 N) dominated the total error of the different force components.

Suggestions for Future Improvement

There are many different ways to improve this lab in the future. First, the inherent systematic error should be the focus of reduction. By using an improved load cell and Baratron, the data collected would be significantly more precise. Additionally, increasing the automation of the lab would also contribute to a great leap in result accuracy. Students should not be

responsible for identifying the increments in which to change the angle of attack. This should be an automated process to ensure that each angle of attack is able to collect three seconds worth of data. Finally, the precise turning of the wing also needs improvement. As seen in the data collected, the wing did not turn by a precise 1° each time. This should just be an easy recalibration of the stepper motor.